

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T23 · Algebraic Proof

With Answers

Questions with full worked solutions

■ 20 pages

FAMILY MAP

Algebraic Proof

F01	Construct an algebraic identity proof	3 Q 8 marks
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F02	Prove divisibility or a remainder	2 Q 6 marks
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F03	Prove a universal parity statement	4 Q 13 marks
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Question 19

4MA1-2H Q14

ORIGINAL QUESTION

3 marks

- 14 (a) Show that $3y(2y + 5)(y + 7)$ can be written in the form $ay^3 + by^2 + cy$ where a, b and c are integers.

(3)

Worked Solution 19

Expansion

MODULAR UNIT 2

3 marks

Expand the two brackets

$$(2y + 5)(y + 7) = 2y^2 + 19y + 35.$$

Multiply by the outside factor

$$3y(2y^2 + 19y + 35) = 6y^3 + 57y^2 + 105y.$$

Final answer

$$6y^3 + 57y^2 + 105y$$

Original question 30

4MA1-2H Q18

ORIGINAL QUESTION

3 marks

- 18 Prove that when the sum of the squares of any two consecutive odd numbers is divided by 8, the remainder is always 2
Show clear algebraic working.

Worked solution

Family: Prove divisibility or a remainder • Variant: Fixed remainder modulo a divisor

METHOD / WORKING

MODULAR UNIT 2 • JANUARY 2022 DETAILED SOLUTIONS
Worked solution 30 MODULAR UNIT 2
Chapter: Algebraic Proof 3 marks
Q18 – Chapter: Algebraic Proof
Family: Prove divisibility or a remainder • Variant: Fixed remainder modulo a divisor
Method
Let the consecutive odd numbers be $2n-1$ and $2n+1$.
Why: Any two consecutive odd numbers have this form.
Method
<i>Theirsquaresum is</i> $(2n-1)^2 + (2n+1)^2 = 8n^2 + 2$.
Why: Expand and add.
Method
$8n^2$ is divisible by 8, so the remainder is always 2.
Why: Separate the multiple of 8 from the remainder .
Final answer
remainder: 2

FINAL ANSWER

- 17 Prove that the difference between two consecutive square numbers is always an odd number.
Show clear algebraic working.

Worked solution

January 2020 | Question 17 | 3 marks

Family: Prove a universal parity statement | Variant: Consecutive integer parity identity

METHOD / WORKING

1. Let consecutive squares be n^2 and $(n+1)^2$.2. Difference = $(n+1)^2 - n^2 = 2n + 1$.3. $2n+1$ is always odd for integer n .

FINAL ANSWER

 $(n+1)^2 - n^2 = 2n + 1$, which is odd

- 17 Using algebra, prove that, given any 3 consecutive even numbers, the difference between the square of the largest number and the square of the smallest number is always 8 times the middle number.

Worked solution

June 2021 | Question 17 | 3 marks

Family: Prove a universal parity statement • Variant: Consecutive even/odd sequence proof

Method / working

1. Represent consecutive even numbers as $2n$, $2n+2$, $2n+4$.
2. Subtract squares: $(2n+4)^2 - (2n)^2 = 16n+16$.
3. Factor: $16n+16 = 8(2n+2)$, which proves the result.

Final answer: Proof: difference = $8(2n+2)$

19 $ABCED$ is a five-sided shape.

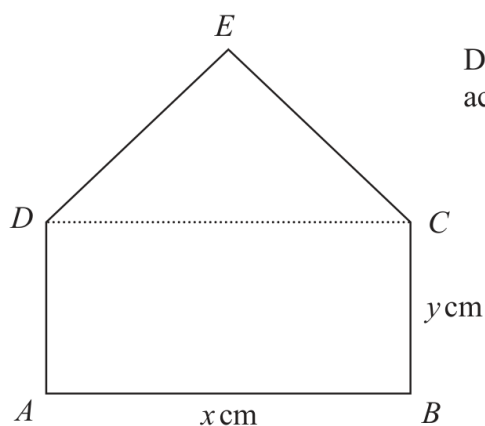


Diagram **NOT**
accurately drawn

$ABCD$ is a rectangle.

CED is an equilateral triangle.

$$AB = x \text{ cm} \quad BC = y \text{ cm}$$

The perimeter of $ABCED$ is 100 cm.

The area of $ABCED$ is $R \text{ cm}^2$

(a) Show that $R = \frac{x}{4} (200 - [6 - \sqrt{3}]x)$

(3)

(b) (i) Find the value of x for which R has its maximum value.

Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$$x = \dots\dots\dots$$

(2)

(ii) Explain why the maximum value of R is given by this value of x .

(1)

Worked solution

November 2021 | Question 19 | 3 marks

Family: Construct an algebraic identity proof | Variant: Derive a general geometric algebra formula

METHOD / WORKING

1. The perimeter is made from three sides of length x and two sides of length y :

2. $3x+2y=100$

3. So

4. $y = \frac{100-3x}{2}$

5. The area is the area of the rectangle plus the area of the equilateral triangle:

6. $R = xy + \frac{\sqrt{3}}{4}x^2$

7. Substitute $y = \frac{100-3x}{2}$:

8. $R = x(\frac{100-3x}{2}) + \frac{\sqrt{3}}{4}x^2$

9. $R = 50x - \frac{3}{2}x^2 + \frac{\sqrt{3}}{4}x^2$

10. $R = \frac{x}{4}(200 - (6 - \sqrt{3})x)$

11. For the maximum, write

12. $R = -\frac{6-\sqrt{3}}{4}x^2 + 50x$

13. The maximum occurs at

14. $x = \frac{-b}{2a} = \frac{-50}{2(-\frac{6-\sqrt{3}}{4})}$

15. $x = \frac{100}{6-\sqrt{3}}$

16. The coefficient of x^2 is negative, so the quadratic graph opens downwards and this vertex gives a maximum.17. Answer: $R = \frac{x}{4}(200 - (6 - \sqrt{3})x)$, and $x = \frac{100}{6-\sqrt{3}}$.

FINAL ANSWER

$$R = \frac{x}{4}(200 - (6 - \sqrt{3})x), \text{ and } x = \frac{100}{6-\sqrt{3}}$$

- 17 Given that $8\sqrt{m} + \sqrt{49m} - \sqrt{9m} = k\sqrt{m}$
where k is an integer and m is a prime number,
(a) work out the value of k

$$k = \dots\dots\dots (1)$$

- (b) Show that $\frac{5 - \sqrt{18}}{1 - \sqrt{2}}$ can be written in the form $a + b\sqrt{2}$
where a and b are integers.
Show each stage of your working clearly.

(3)

(Total for Question 17 is 4 marks)

Answer reference | January 2023 | Question 17

Family: Prove a universal parity statement • Variant: Product or sum of arbitrary odd/even numbers

Question	Working	Answer	Mark	Notes
17 (a)		12	1	B1
(b)	$\frac{5-\sqrt{18}}{1-\sqrt{2}} \times \frac{1+\sqrt{2}}{1+\sqrt{2}} \text{ or } \frac{5-\sqrt{18}}{1-\sqrt{2}} \times \frac{-1-\sqrt{2}}{-1-\sqrt{2}} \text{ oe}$		3	M1 Multiplying numerator and denominator by $1+\sqrt{2}$
	$\frac{5-\sqrt{36}+5\sqrt{2}-\sqrt{18}}{1+\sqrt{2}-\sqrt{2}-2} \text{ or } \frac{5-6-3\sqrt{2}+5\sqrt{2}}{-1} \text{ or}$ $\frac{-5+6+3\sqrt{2}-5\sqrt{2}}{1} \text{ oe}$ NB: allow $\sqrt{18}$ or $3\sqrt{2}$ $\sqrt{36}$ or 6 or $\sqrt{6}\sqrt{6}$			M1 Showing correct expansions (not necessarily as a fraction)
	working required	$1-2\sqrt{2}$		A1 dep on M2 (ie all stages of working must be shown convincingly) or for stating $a = 1$ and $b = -2$
Total 4 marks				

Worked solution

January 2023 | Question 17 | 4 marks

Family: Prove a universal parity statement • Variant: Product or sum of arbitrary odd/even numbers

Official final mark-scheme pages are embedded immediately after this question.

5 Show that $3\frac{5}{7} + 1\frac{2}{3} = 5\frac{8}{21}$

(Total for Question 5 is 3 marks)

Worked solution

November 2025 | Question 5 | 3 marks

Family: Prove divisibility or a remainder • Variant: Expression built from a known multiple

Solution

$\frac{37}{21} + \frac{13}{21}$

Use denominator 21:

$\frac{37}{21} = \frac{9 \times 21}{21}$, and $\frac{13}{21} = \frac{7 \times 21}{21}$

$\frac{9 \times 21}{21} + \frac{7 \times 21}{21} = \frac{16 \times 21}{21}$

Answer: $\frac{16 \times 21}{21}$.

PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

17 Show that $\frac{2\sqrt{7} + 2}{\sqrt{7} - 3}$ can be written in the form $a - \sqrt{b}$ where a and b are integers.

Show your working clearly.

(Total for Question 17 is 3 marks)

Worked solution

November 2025 | Question 17 | 3 marks

Family: Prove a universal parity statement • Variant: Product or sum of arbitrary odd/even numbers

Solution

$$(2\sqrt{7}+2)/(\sqrt{7}-3) \times (\sqrt{7}+3)/(\sqrt{7}+3)$$

$$=((2\sqrt{7}+2)(\sqrt{7}+3))/(7-9)$$

$$=(14+6\sqrt{7}+2\sqrt{7}+6)/(-2)$$

$$=(20+8\sqrt{7})/(-2)=-10-4\sqrt{7}$$

Since $4\sqrt{7}=\sqrt{112}$,

$$-10-4\sqrt{7}=-10-\sqrt{112}$$

Answer: $-10-\sqrt{112}$, so $a=-10$ and $b=112$.

16 Use algebra to show that $0.3\dot{7}\dot{8} = \frac{25}{66}$

(Total for Question 16 is 2 marks)

Worked solution

November 2025 | Question 16 | 2 marks

Family: Construct an algebraic identity proof | Variant: Derive a general geometric algebra formula

METHOD / WORKING

Worked solution
November 2025 Question 16 2 marks
Family: Construct an algebraic identity proof Variant: Derive a general geometric algebra formula
Solution
Let
$x=0.3787878\dots$
Then
$10x=3.787878\dots$
and
$1000x=378.787878\dots$
Subtract:
$990x=375$
$x=(375)/(990)=(25)/(66)$
Answer : $0.3787878\dots = d_{66}^{25}$.
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER